



Decoding Market Movements: PCA-Enhanced Machine Learning for S&P 500 Log Return Prediction

Design of research

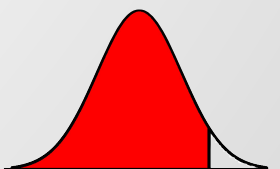
Student Name : Katherine Liu

Advisor Name : Hui Wen Teng

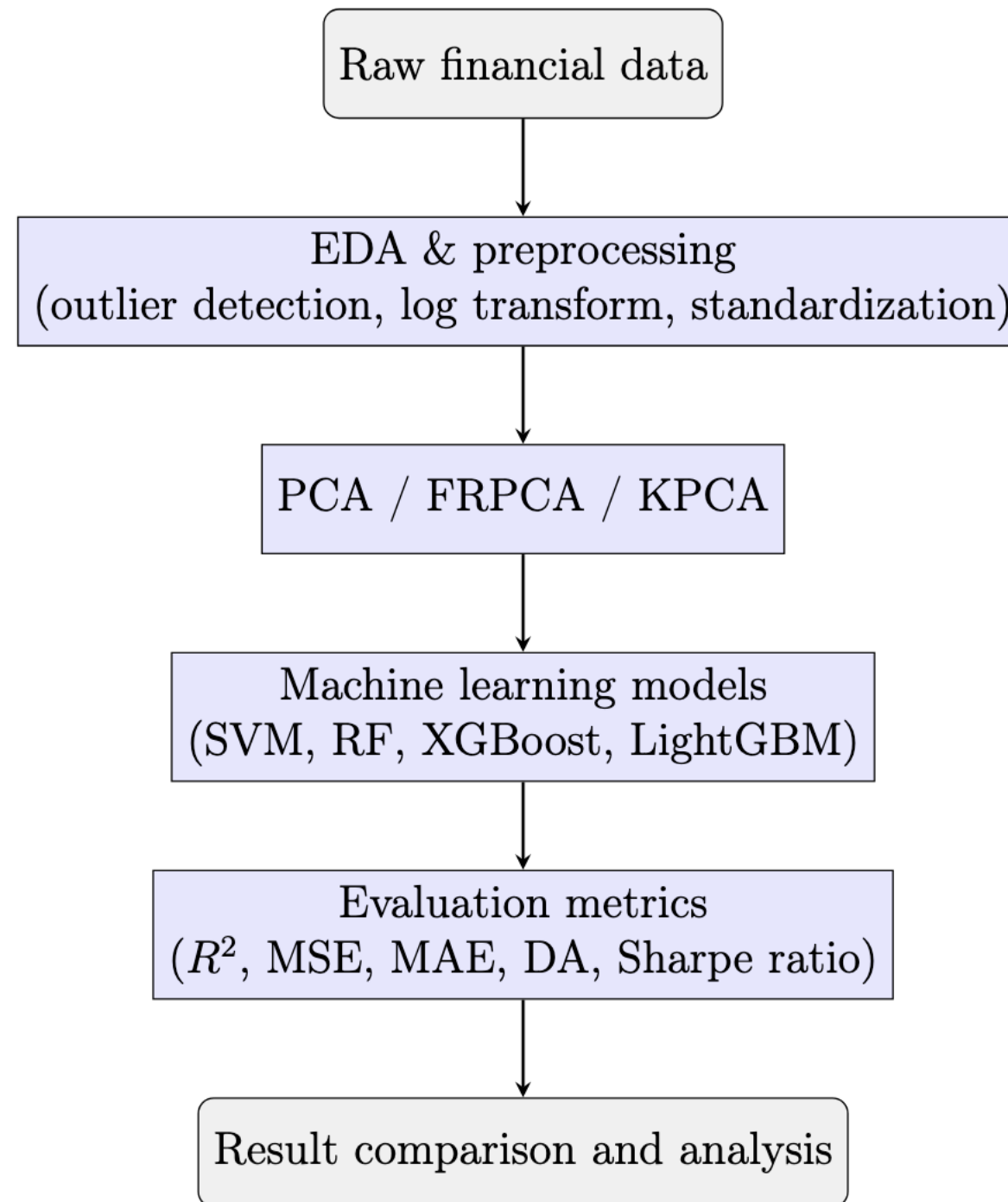
National Yang Ming Chiao Tung University
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Outline

1. Overall research framework
2. Dimensionality reduction methods
3. Machine learning models for prediction
4. Performance measures



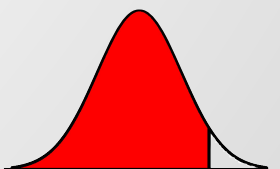
Overall research framework



Dimensionality reduction methods

▣ PCA

- ▶ Reduce high-dimensional data to lower dimensions
- ▶ Preserve variance (more information)
- ▶ Identify main directions of variation → project onto top principal components



Dimensionality reduction methods

▣ FRPCA

- ▶ Robust to outliers and noise
- ▶ Decompose data into main structure and outliers (noise)
- ▶ Apply PCA on the low-rank matrix → obtain robust principal components



Dimensionality reduction methods

▣ KPCA

- ▶ Capture nonlinear relationships in the data
- ▶ Map data to a high-dimensional feature space, then apply PCA
- ▶ Extract nonlinear patterns → better for complex datasets



Machine learning models for prediction

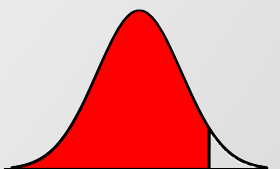
- Support Vector Regression (SVR / SVM)
 - ▶ Capture nonlinear relationships in the data
 - ▶ Handles nonlinearity via kernel functions
 - ▶ Robust to small deviations and avoids overfitting



Machine learning models for prediction

▣ Random Forest (RF)

- ▶ Ensemble of decision trees, averaging predictions for stability
- ▶ Captures complex nonlinear patterns and feature interactions
- ▶ Resistant to overfitting due to randomness in samples and features



Machine learning models for prediction

- ▣ Extreme Gradient Boosting (XGBoost)
 - ▶ Sequentially builds trees to correct residual errors of previous trees
 - ▶ Includes regularization to improve generalization
 - ▶ Efficient for high-dimensional datasets, captures nonlinear relationships



Machine learning models for prediction

- ▣ Light Gradient Boosting Machine (LightGBM)
 - ▶ Builds trees leaf-wise for faster convergence and lower loss
 - ▶ Efficient for large datasets with high-dimensional features
 - ▶ Captures complex patterns and interactions while maintaining speed



Performance measures

- ▣ R^2

- ▶ how well predictions explain the variance in actual data

- ▣ MSE

- ▶ Average of squared differences between predicted and actual values

- ▣ MAE

- ▶ Average of absolute differences between predicted and actual values

- ▣ Directional Accuracy (DA)

- ▶ Proportion of predictions with the correct up/down direction

- ▣ Sharpe Ratio

- ▶ Risk-adjusted return based on model predictions





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